

ECN 1620 Prices, Markets and Money: formula sheet

Version 2025.1. Issued Monday 15 September 2025. Prof. T. Kessandru.

This sheet is printed on the back page of the midterm and final exam papers. You may not bring your own copy into the exam room.

A. Elasticity

- Percentage change = $(\text{new} - \text{old}) / \text{old} \times 100$. Use the starting value as the base.
- Price elasticity of demand (PED) = $\% \text{ change in quantity demanded} / \% \text{ change in price}$. Keep the minus sign.
- Elastic: PED below -1. Unit elastic: PED equal to -1. Inelastic: PED between -1 and 0.
- Cross-price elasticity = $\% \text{ change in quantity of good A} / \% \text{ change in price of good B}$. Positive: substitutes. Negative: complements.
- Income elasticity = $\% \text{ change in quantity} / \% \text{ change in income}$. Negative: inferior good. Between 0 and 1: necessity. Above 1: luxury.
- Price elasticity of supply = $\% \text{ change in quantity supplied} / \% \text{ change in price}$.
- Total revenue = price $\times$ quantity.

B. Prices

- CPI in year t = cost of the basket in year t / cost of the basket in the base year, $\times 100$.
- Inflation rate = $(\text{CPI this year} - \text{CPI last year}) / \text{CPI last year} \times 100$.
- Real rate = nominal rate - inflation, when inflation is below 10%. At 10% or more: $(1 + \text{nominal}) = (1 + \text{real}) \times (1 + \text{inflation})$.
- Real value = nominal value / CPI, $\times 100$.

C. Money and policy

- Quantity equation: $M \times V = P \times Y$. Growth form: money growth + velocity growth = inflation + real output growth.
- Money multiplier = $1 / \text{reserve ratio}$.
- Taylor rule: policy rate = neutral real rate + inflation + $0.5 \times (\text{inflation} - \text{target}) + 0.5 \times \text{output gap}$. Use a neutral real rate of 2% and a target of 2% unless told otherwise.

D. Rounding

- All results: 1 decimal place.
- Round only at the end.

E. Symbols

- P price, Q quantity, TR total revenue.
- M money supply, V velocity of circulation, Y real output.
- CPI consumer price index, with the base year set to 100.
- The output gap is actual output less potential output, as a percentage of potential. A negative gap means the economy is producing below potential.

F. Reminders

- State the method in one line before you calculate. A bare number earns no method marks.
- Give units. An interest rate is in percent a year; an elasticity has no units.